16 The curve **C** has equation $y = x^2 + 3x - 3$

The line **L** has equation $y - 5x + 4 = 0$

Show, algebraically, that **C** and **L** have exactly one point in common.

Line L: $y = 5x - 4$

At a common point: $x^2 + 3x - 3 = 5x - 4$

Rearrange: $x^2 - 2x + 1 = 0$

Factorise: $(x - 1)^2 = 0$, so $x = 1$ only.

Then $y = 1$; exactly one intersection $(1,1)$.

1 mark
eliminate y

1 mark
quadratic equals zero

1 mark
$(x-1)^2=0$

1 mark
one point in common

(Total for Question 16 is 4 marks)

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